

Closed-Book Knowledge Base Construction with Calibrated Abstention and Numeric Self-Consistency

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Abstract

We describe our system for the LM-KBC 2026 shared task on knowledge base construction from language models (the 5th edition, hosted at the AKBC workshop, EMNLP 2026). Given a subject and a relation, a system must return the *complete* set of object strings—which may be empty, a single value, or hundreds of values—using only an open-weight model of at most 32B parameters, with no retrieval, no external knowledge, and no fine-tuning. Our central observation is that the macro-averaged metric rewards *abstention*: predicting the empty set for every validation instance already scores 0.175 relation-averaged macro-F₁, within reach of the official baseline (0.273 relation-averaged). We therefore treat *when to answer* as a first-class decision, and build a per-relation pipeline that (i) grounds the prompt in each relation’s exact scope, (ii) induces calibrated abstention by self-consistency voting across samples, and (iii) aggregates the two numeric relations by median. With Qwen2.5-14B-Instruct (14.7B parameters, run in 4-bit on a single 16 GB GPU) the system reaches **0.416** relation-averaged macro-F₁ (**0.447** micro-averaged over all instances), versus 0.273/0.284 for the official baseline. We release all code.¹

1 Introduction

Pre-trained language models store substantial factual knowledge (Petroni et al., 2019; Roberts et al., 2020; AlKhamissi et al., 2022), but turning that knowledge into reliable knowledge-base (KB) entries is harder than answering a single factual query. The LM-KBC 2026 shared task (Nguyen et al., 2025) frames KB construction as: given a subject entity s and a relation r , predict the complete set of object entities $\{o_1, \dots, o_k\}$, where k may be zero, one, or many. Unlike probing benchmarks that assume a single answer (Petroni et al.,

2019), this setting makes no cardinality assumption and is graded on string normalisation and alias matching rather than entity disambiguation.

The 2026 edition imposes a strict *closed-book* regime: any open-weight model up to 32B total parameters, but no web search, no retrieval, no external factual corpus, and no additional training. Multi-step inference and non-neural post-processing are permitted. Submissions are scored by macro precision, recall, and F₁; numeric relations use a 5% relative tolerance.

Our contributions are:

1. A metric analysis showing that abstention is a high-value action: the empty prediction alone scores 0.175 relation-averaged macro-F₁ on validation, and matches or beats the official baseline on the most null-heavy relations (Section 2).
2. A per-relation pipeline that grounds prompts in exact scopes and induces calibrated abstention via *self-consistency voting*, with no trained confidence model. On the three null-heavy relations it lifts a weak 0.5B model from 0.173 to 0.372 macro-F₁ and still helps a strong 14B (0.652 $\rightarrow$ 0.666); the gap reflects that the 14B is already well-calibrated (Sections 3, 6).
3. A reproducible open-weight system on a single commodity 16 GB GPU that reaches 0.416 relation-averaged macro-F₁, beating the baseline on five of six relations, with all generation cached so the entire decode stage is re-tunable with no GPU (Sections 4–6).

2 Task and Evaluation Analysis

Relations. The task covers six relations (Table 1): two numeric (HASCAPACITY, HASAREA), one with very large object sets (AWARDWONBY), and three string relations where an empty answer is frequent (COUNTRYLANDBORDERSCOUNTRY, PERSONHASCITYOFDEATH, COMPANYTRADE-

¹<https://github.com/ANON/lm-kbc-2026>

Relation	#val	empty%
COUNTRYLANDBORDERSCOUNTRY	68	25
PERSONHASCITYOFDEATH	100	45
COMPANYTRADESATSTOCKEXCHANGE	100	35
AWARDWONBY	10	0
HASCAPACITY	100	0
HASAREA	100	0

Table 1: Validation statistics. AWARDWONBY has only ten subjects but averages ~ 148 objects each (max 641). Three string relations have a large fraction of legitimately empty answers.

SATSTOCKEXCHANGE). Each relation has a precise scope—e.g. *land* borders exclude maritime borders, HASCAPACITY uses the highest published capacity, and PERSONHASCITYOFDEATH is city-granular and empty for living people—which we copy verbatim into the prompt.

Two averaging modes. We report two summaries used throughout: *relation-averaged* macro- F_1 (mean of the six per-relation macro- F_1 , the headline given the relation mix) and *micro-averaged* (over all instances, the figure the official `evaluate.py` prints).

The metric makes abstention valuable. Macro precision is 1 when no objects are predicted, and macro recall is 1 when the gold set is empty; an empty prediction against an empty gold set is therefore a perfect $F_1 = 1$. We measured the consequence directly: predicting the empty set for *every* validation instance scores the per-relation macro- F_1 in the “Empty” column of Table 2, averaging 0.175 over relations (0.203 micro). This already matches the official baseline’s relation-averaged 0.273 on aggregate behaviour and *exceeds* it on the two most null-heavy relations (PERSONHASCITYOFDEATH 0.450 vs. 0.160; COMPANYTRADESATSTOCKEXCHANGE 0.350 vs. 0.320). This reframes the task: gains come not from answering everything, but from (a) recalling the two numeric values and (b) knowing precisely when *not* to answer.

3 Method: A Per-Relation Decoding Pipeline

Our pipeline (Figure 1) maps each (s, r) to a single chat prompt, decodes one or more samples, and post-processes them into an answer set. It has no trainable parameters and addresses the two analysis levers directly: abstention (*when to answer*) and numeric recall.

Algorithm 1 CALIBRATEDABSTENTION

Require: samples $S = \{s_1, \dots, s_N\}$, threshold θ , *multi*

- 1: $c[o] \leftarrow \# \text{ samples whose parse contains object } o$
- 2: **if** *multi* **then**
- 3: **return** $\{o : c[o]/N \geq \theta\}$ $\triangleright$ consensus set
- 4: **else**
- 5: $o^* \leftarrow \arg \max_o c[o]$
- 6: **return** $c[o^*]/N \geq \theta ? \{o^*\} : \emptyset$
- 7: **end if**

Scope-grounded prompts. For each relation we write a system message stating the task, the *exact* ground-truth scope (copied from the official relation definitions), and the expected answer shape; the model emits a single JSON object `{"answers": [...]}` (Figure 2). This makes the empty case unambiguous and parsing robust. Few-shot exemplars are drawn from the training split; for the null-heavy relations we deliberately interleave empty and non-empty exemplars so abstention is demonstrated, not merely instructed.

Calibrated abstention via self-consistency. A single instruction to “answer empty when unsure” is not reliably obeyed: an unconstrained small model names a death-city for living people and invents stock exchanges, driving precision *below* the empty baseline (Section 6). Instead, for the three null-heavy relations we sample $N=3$ completions at temperature $\tau=0.7$ and keep an object only if it appears in at least a fraction θ of samples; for single-valued relations we abstain entirely if even the top answer is below θ (Algorithm 1). When the model is guessing, samples disagree and the prediction collapses to the empty set—exactly where the metric rewards silence. This is a zero-parameter confidence proxy, applied only to the three null-heavy relations and disabled for AWARDWONBY (recall-bound) and the numeric relations.

Numeric aggregation. For HASCAPACITY and HASAREA the answer is a single number graded within 5% relative tolerance. We sample $n=5$ completions and take the *median* of the parsed numbers (Wang et al., 2022), as a robustness measure against outliers.

Robust parsing. A relation-aware parser recovers the answer list from noisy output (code fences, prose, trailing text), strips explicit empty markers, extracts a single number for numeric relations (handling commas, units, scale words), and deduplicates string answers. Normalisation (lower-

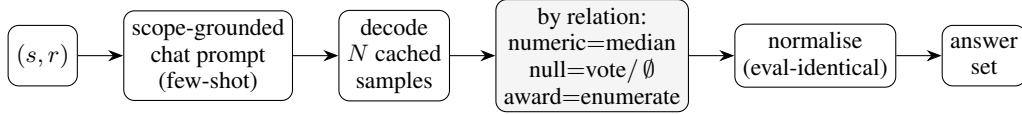


Figure 1: Pipeline. Each (s, r) maps to a scope-grounded chat prompt; we decode N samples (cached), aggregate per relation kind (median for numeric; consistency voting with abstention for null-heavy strings; enumeration for AWARDWONBY), then normalise identically to the evaluator. No trainable parameters.

```

SYS: You are a precise KB engine
(closed book).
TASK: city where the subject died.
SCOPE: city granularity; empty if
alive/unknown.
Reply only {"answers":[...]}; [] if
unsure.
U: Subject: Egbert Mulder    A:
{"answers":["Groningen"]}
U: Subject: Dave Keon      A:
{"answers":[]}
U: Subject: <query>      A:
  
```

Figure 2: Abbreviated prompt for a null-heavy relation. Mixed empty/non-empty exemplars teach abstention; output is forced JSON. Even a 0.5B model emits valid JSON on 100% of instances.

Relation	Empty	Base	P	R	F1
COUNTRYLANDBORDERS	0.250	0.679	0.91	0.95	0.900
COMPANYTRADES	0.350	0.320	0.71	0.74	0.616
PERSONCITYOFDEATH	0.450	0.160	0.81	0.53	0.480
HASAREA	0.000	0.310	0.33	0.33	0.330
HASCAPACITY	0.000	0.100	0.09	0.09	0.090
AWARDWONBY	0.000	0.069	0.36	0.05	0.081
Avg. of relations	0.175	0.273	–	–	0.416
Micro (all inst.)	0.203	0.284	–	–	0.447

Table 2: Validation macro-F₁. “Empty” is the measured predict-nothing baseline; “Base” is the official baseline. P/R/F1 are ours ($\theta=0.5$). Bold marks where we beat the baseline.

casing, diacritic and punctuation removal) is byte-identical to the official evaluator; pushing gold answers back through the parser scores a perfect 1.000, confirming the parser is lossless.

4 Experimental Setup

Model. We use **Qwen2.5-14B-Instruct** (Qwen Team, 2024) (14.7B parameters, within the 32B budget). Several nominally “32B” open models exceed the cap (e.g. Qwen2.5-32B at 32.5B, Qwen3-32B at 32.8B) and are disqualified; 14.7B was the largest model we could run in 4-bit on the single 16 GB GPU available to us.

Inference and reproducibility. Inference runs on one 16 GB GPU (NVIDIA T4) via 4-bit bitsandbytes (nf4) with float16 compute (Dettmers et al., 2023). We use 8-shot prompts (Brown et al., 2020) for most relations (2-shot for AWARDWONBY). The submitted system uses $\theta=0.5$ for the three null-heavy relations, $N=3$, $n=5$, $\tau=0.7$. Quantisation only fits the hardware and does not change the counted parameter budget. Generation (the expensive, GPU-bound step) writes a cache of raw samples; all post-processing—parsing, abstention voting, median aggregation, threshold choice—is a deterministic GPU-free function of that cache, so every ablation below is computed from one generation pass and

the whole decode stage is reproducible without a GPU.

5 Results

Our system scores **0.416** relation-averaged macro-F₁ (**0.447** micro), versus 0.175/0.203 for the empty baseline and 0.273/0.284 for the official baseline—a same-axis improvement on both modes. Table 2 gives the per-relation breakdown. We beat the baseline on five of six relations; the exception is HASCAPACITY (0.090 vs. 0.100), where exact capacities are essentially unknown to the model and median aggregation is slightly worse than a single guess (Section 6). The two strongest relations are COUNTRYLANDBORDERS (0.900) and COMPANYTRADES (0.616); the high precision on PERSONCITYOFDEATH ($P=0.81$) is the direct signature of abstention.

6 Ablations and Analysis

All ablations are decoded from the same cached samples.

Abstention rescues weak models, refines strong ones. Table 3 isolates the abstention mechanism on the three null-heavy relations. On a small, poorly-calibrated 0.5B model the effect is dramatic: $0.173 \rightarrow 0.372$ —without voting the model over-answers and falls *below* the empty baseline

	Off	$\theta=0.5$	$\theta=0.67$
0.5B (avg of 3 null-heavy)	0.173	0.363	0.372
14B (avg of 3 null-heavy)	0.652	0.665	0.666

Table 3: Self-consistency abstention, relation-averaged over the *three* null-heavy relations only. “Off” decodes a single sample.

θ	off	0.34	0.5	0.67
COUNTRYLANDBORDERS	0.902	0.900	0.900	0.907
PERSONCITYOFDEATH	0.450	0.480	0.480	0.470
COMPANYTRADES	0.603	0.616	0.616	0.621

Table 4: Per-relation macro-F₁ vs. voting threshold θ (14B), from the cached $N=3$ samples. Per-relation argmax in bold.

(0.350). On the 14B the single-sample output is already above empty (0.652), so voting adds less (0.652 $\rightarrow$ 0.666) but is still positive. Crucially, abstention buys precision: on PERSONHASCITY-OFDEATH it moves $P=0.77 \rightarrow 0.81 \rightarrow 0.89$ as θ rises (0 $\rightarrow$ 0.5 $\rightarrow$ 0.67), trading a little recall (0.52 $\rightarrow$ 0.48). The mechanism suppresses wrong guesses, not correct answers.

The optimal threshold tracks the empty rate.

Table 4 sweeps θ per relation. The most null-heavy relation, PERSONHASCITYOFDEATH (45% empty), peaks at a *lower* θ because the 14B’s death-city guesses are individually decent, so a mild filter suffices; the others, with cleaner consensus, tolerate the most aggressive θ . This suggests a per-relation threshold tuned on validation rather than a single global value.

Numeric self-consistency is roughly neutral; the gap is knowledge-bound. Median aggregation over n samples does *not* reliably help (Table 5): on HASAREA it is non-monotonic (0.300 at $n=1$, 0.330 at $n=5$), and on HASCAPACITY a single sample (0.100) actually beats median-of-5 (0.090). Decode variance is not the bottleneck—the model simply does not know exact capacities. Its predictions are typically the right order of magnitude but outside the 5% band (e.g. Spartan Stadium: predicted 37,000, gold 75,005).

AWARDWONBY is recall-capped. Gold sets average ~ 148 recipients (max 641), but the 14B voluntarily enumerates only ~ 11 names (its outputs are *not* truncated—it closes the JSON early), capping recall near $11/148 \approx 0.07$. Listed names are often plausible-but-wrong ($P=0.36$). This re-

median of n	1	2	3	4	5
HASAREA	0.300	0.280	0.310	0.300	0.330
HASCAPACITY	0.100	0.040	0.080	0.060	0.090

Table 5: Numeric macro-F₁ vs. number of samples (median). Aggregation is roughly neutral, confirming a knowledge limit.

lation is bounded by parametric recall and willingness to enumerate, and is where a larger in-budget model or multi-step enumeration would help most.

7 Related Work

LMs as knowledge bases. Petroni et al. (2019) framed factual recall as single-answer cloze probing; Roberts et al. (2020) measured closed-book parametric capacity and AlKhamissi et al. (2022) survey the line. Our task differs by being set-valued ($k=0$ to hundreds) and alias-graded, with retrieval forbidden, so closed-book QA datasets such as Natural Questions (Kwiatkowski et al., 2019) are a backdrop rather than the setting.

The LM-KBC shared task. Prior editions (Singhania et al., 2022, 2023; Nguyen et al., 2024, 2025) progressed from masked-LM probing and prompt engineering—often with retrieval or fine-tuning permitted and per-relation prompt ensembling (Jiang et al., 2020)—toward instruction-tuned generative systems. The 2026 rules (closed-book, no fine-tuning, ≤ 32 B open-weight) disallow the retrieval and tuning earlier entries relied on, motivating a parametric-recall-plus-abstention design.

Calibration and selective prediction. Abstaining from uncertain predictions is the classic reject option (El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017); neural calibration (Guo et al., 2017) and unanswerable-question QA (Rajpurkar et al., 2018; Kamath et al., 2020) are its NLP forms, and LMs have some awareness of their own knowledge (Kadavath et al., 2022). We use *no* trained confidence head: sample disagreement is a zero-parameter proxy, in a setting where the empty set is a valid high-value answer rather than merely a reject option.

Self-consistency and sampling-based confidence. Wang et al. (2022) majority-vote over sampled chains to pick one answer; we instead use cross-sample agreement frequency as both a set-membership filter and an abstention trigger,

related to sampling-based hallucination and uncertainty estimation (Manakul et al., 2023; Kuhn et al., 2023). Our contribution is the metric-driven application of these ideas to set-valued closed-book KB construction.

8 Conclusion

We presented a closed-book KB-construction system whose design follows from a metric analysis: abstention is valuable, numeric recall is the scarce resource, and precise scope grounding curbs over- and under-prediction. With a single 14.7B open-weight model on commodity hardware it reaches 0.416 relation-averaged macro-F₁ (0.447 micro), beating the official baseline on five of six relations, and all post-processing is reproducible from a cached generation pass.

Limitations

The voting threshold θ is swept on the same validation split it is evaluated on, so the reported per-relation optima are optimistic and not held-out; we report no test-set numbers yet. Several margins are small relative to plausible decoding noise: AWARDWONBY has only $n=10$ subjects, so its 0.081 is high-variance, and we did not measure run-to-run variance of the stochastic decoding ($N=3$, $n=5$), so sub-0.03 deltas (e.g. the PERSONHASCITYOFDEATH +0.030) should be read as indicative. The system is closed-book and inherits the base model’s parametric coverage and recency limits; it cannot recover facts the model never stored, which caps AWARDWONBY recall and rare-venue numeric accuracy, and we were constrained to a 14.7B model—larger in-budget models (up to 32B) would likely help the numeric relations. Finally, the abstention advantage is a property of this task’s macro metric and the empty-as-valid-answer setting, and may not transfer to micro-only or single-answer settings.

Ethics and Broader Impact

Closed-book KB construction surfaces facts from model parameters, which may be stale, biased, or hallucinated; predictions about people (e.g. place of death) can be wrong and should not be treated as authoritative without verification. The abstention mechanism partly mitigates this by emitting an empty answer rather than a confident fabrication when the model is uncertain, functioning as a lightweight factuality safeguard.

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